

0 3 . 1

Describe how we breathe in.

[3 marks]

0 3 . 2

A scientist prepared alveolar tissue to view using an optical microscope. The scientist cut very thin slices of the alveolar tissue.

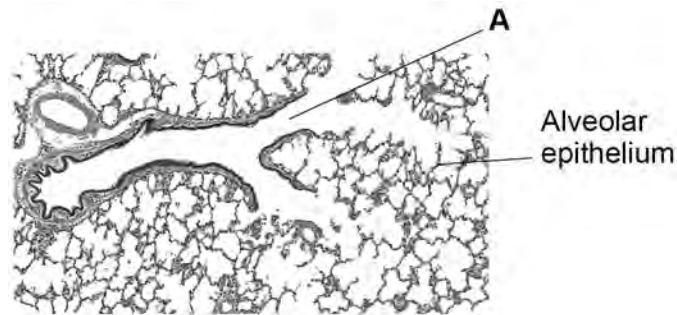
Explain why the scientist used very thin slices of alveolar tissue with the optical microscope.

[2 marks]

0 3 . 3

Figure 2 is an image of the lung tissue observed using an optical microscope.

Figure 2



Identify the tube labelled **A**.

[1 mark]

A _____

The scientist used a ruler to measure the diameter of some of the alveoli.

Table 2 shows the scientist's results.

Table 2

Alveolus diameter / mm								
Alveolus diameter / mm	4	2	5	1	2	3	5	2

0 3 . 4

The magnification of the image in **Figure 2** is $\times 40$

Use this information and **Table 2** to calculate the mean diameter, in μm , of the alveoli.

Show your working.

[2 marks]

Answer _____ μm

0	3	.	5
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Give the uncertainty associated with taking a measurement using a ruler with 1 mm graduations.

Calculate the percentage error for a measurement using the ruler of 4 mm

[2 marks]

Uncertainty $\pm$ _____ mm

Percentage error _____

